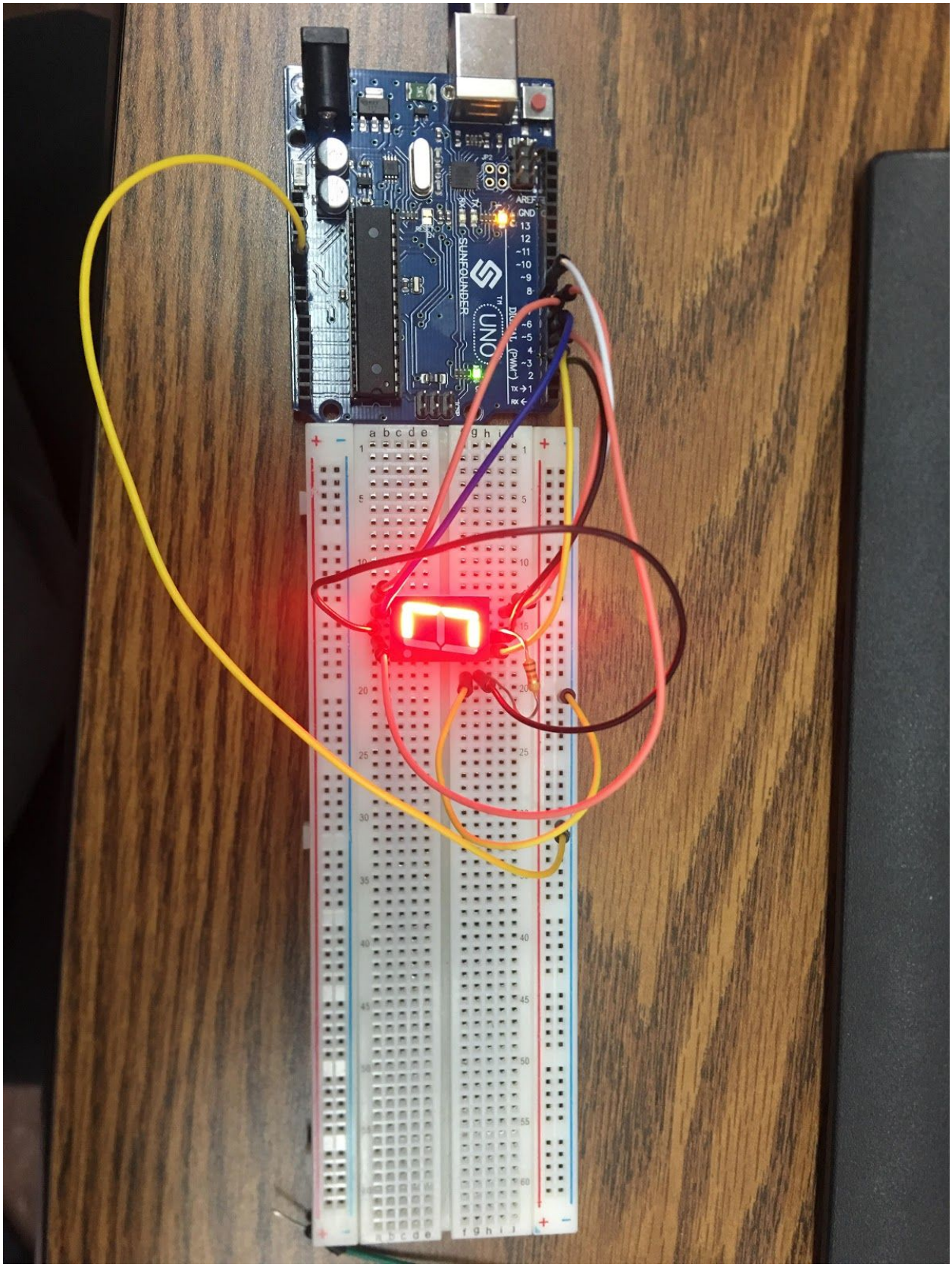
```
How many grades should be averaged?
Enter in a grade (0-100)
Your number average is: 67
How many grades should be averaged?
```

TUTORIAL_-_CASE_STRUCTURES

```
/*
   Title: TUTORIAL - CASE STRUCTURES
   Description: Use case structures to make a seven segment display display the numbers 0-9, then a dash.
   Created By: Doniyor Nimatullo
   Started and Finished : 11/14/18
*/
const int SLA = 2, SLB = 3, SLC = 4, SLD = 5, SLE = 6, SLF = 7, SLG = 8, timedelay = 500;
int predivision = 0, gradenum, numaverage = 0, letter = 0;

void setup() {
  SerialMonitor();
  Initialize();
}

void loop() {
  AskGrades();
}

void Initialize()
{
  pinMode(SLA, OUTPUT);
  pinMode(SLB, OUTPUT);
  pinMode(SLC, OUTPUT);
  pinMode(SLD, OUTPUT);
  pinMode(SLE, OUTPUT);
  pinMode(SLF, OUTPUT);
  pinMode(SLG, OUTPUT);
}

void SerialMonitor()
{
  Serial.begin(9600);
  Serial.flush();
}

void AskGrades()
{
  Serial.println("How many grades should be averaged? ");
  while (Serial.available() == 0) {};
  gradenum = Serial.parseInt(); //sets amount of grades to user input
  CalculateAverage(gradenum);
}

void CalculateAverage(int howmany)
{
  for (int i = 1; i <= howmany; i++) //loops 'gradenum' times
```

```
  {
    Serial.println("Enter in a grade (0-100) ");
    while (Serial.available() == 0) {};
    predivision = predivision + Serial.parseInt(); //sum of grades
    numaverage = predivision / howmany; //average of grades
    Serial.print("Your number average is: ");
    Serial.println(numaverage);
    SevenSegment();
  }
}

void SevenSegment()
{
  switch (numaverage) //variable to check
  {
    case 90 ... 100:
      {
        (1,1,1,0,1,1,1); //letter A
        delay(timedelay);
        break;
      }

    case 80 ... 89: //letter B
      {
        (0,0,1,1,1,1,1);
        delay(timedelay);
        break;
      }

    case 70 ... 79: //letter C
      {
        (1,0,0,1,1,1,0);
        delay(timedelay);
        break;
      }

    case 60 ... 69: //letter D
      {
        (0,1,1,1,1,0,1);
        delay(timedelay);
        break;
      }

```

```
    numaverage = prediction / howmany; //average of grades
    Serial.print("Your number average is: ");
    Serial.println(numaverage);
    SevenSegment();
  }
}

void SevenSegment()
{
  switch (numaverage) //variable to check
  {
    case 90 ... 100:
      {
        (1,1,1,0,1,1,1); //letter A
        delay(timedelay);
        break;
      }

    case 80 ... 89: //letter B
      {
        (0,0,1,1,1,1,1);
        delay(timedelay);
        break;
      }

    case 70 ... 79: //letter C
      {
        (1,0,0,1,1,1,0);
        delay(timedelay);
        break;
      }

    case 60 ... 69: //letter D
      {
        (0,1,1,1,1,0,1);
        delay(timedelay);
        break;
      }

    default: //letter F
      {
        (1,0,0,0,1,1,1);
        delay(timedelay);
        break;
      }
  }
}
```


AREF
GND
13
12
~11
~10
~9
8
~6
~5
4
~3
2
TX→1
RX←
SUNFOUNDER
UNO
a b c d e
f g h i j